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MILITARY SPECIFICATION

DRILLING MACHINE, RADIAL, FLOOR MOUNTED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers floor mounted, self-contained electric motor driven radial drilling machines suitable for general purpose drilling, tapping, boring, facing and reaming operations on large size or heavy workpieces with equipment and accessories specified herein. The requirements set forth in this specification and the completed ordering data (see paragraph 6.2), establishes the Government's minimum requirements for each procurement.

1.2 Classification. The machines covered by this specification are of the plain radial arm type and are of the following sizes. The size furnished shall be as specified (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, 2163 Airways Boulevard, Memphis, Tennessee 38114-5051, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3413

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Size A4 - 9-inch column diameter by 3-feet arm length
Size A6 - 9-inch column diameter by 4-feet arm length
Size B1 - 11-inch column diameter by 3-feet arm length
Size B2 - 11-inch column diameter by 4-feet arm length
Size B3 - 11-inch column diameter by 5-feet arm length
Size C1 - 13-inch column diameter by 4-feet arm length
Size C2 - 13-inch column diameter by 5-feet arm length
Size D1 - 15-inch column diameter by 4-feet arm length
Size D2 - 15-inch column diameter by 5-feet arm length
Size D3 - 15-inch column diameter by 6-feet arm length
Size E1 - 17-inch column diameter by 5-feet arm length
Size E2 - 17-inch column diameter by 6-feet arm length
Size E3 - 17-inch column diameter by 7-feet arm length
Size F1 - 19-inch column diameter by 6-feet arm length
Size F2 - 19-inch column diameter by 7-feet arm length
Size F3 - 19-inch column diameter by 8-feet arm length
Size G1 - 22-inch column diameter by 7-feet arm length
Size G2 - 22-inch column diameter by 8-feet arm length
Size G3 - 22-inch column diameter by 9-feet arm length
Size G4 - 22-inch column diameter by 10-feet arm length
Size H1 - 26-inch column diameter by 8-feet arm length
Size H2 - 26-inch column diameter by 9-feet arm length
Size H3 - 26-inch column diameter by 10-feet arm length
Size H4 - 26-inch column diameter by 11-feet arm length
Size H5 - 26-inch column diameter by 12-feet arm length

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

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MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interferences.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

(Unless otherwise indicated, copies of Federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, (ATTN: Mr. Stephen Lowell), Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

U. S. DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

29 CFR 1910 - Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402-0001.)

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Non-Government publications. The following document form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN GEAR MANUFACTURERS' ASSOCIATION (AGMA)

AGMA 218.01 - Standard for Rating the Pitting Resistance and Bending Strength of Spur and Helical Involute Gear Teeth.

AGMA 390.03a - Gear Classification, Materials and Measuring Methods for Bevel, Hypoid, Fine Pitch Wormgearing and Racks Only as Unassembled Gears.

AGMA 2000-A88 - Gear Classification and Inspection Handbook.

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(Application for copies should be addressed to the American Gear Manufacturers' Association, Standards Department, 1500 King Street, Suite 201, Alexandria, VA 22314-2717.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/NFPA 79 - Electrical Standards for Industrial Machinery. (DOD adopted)

ANSI B5.1 - T-Slots, Their Bolts, Nuts and Tongues. (DOD adopted)

ANSI B5.10 - Machine Tapers. (DOD adopted)

ANSI B11.8 - Drilling, Milling, and Boring Machines, Safety Requirements for the Construction, Care, and Use. (DOD adopted)

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Department, 1430 Broadway, New York, NY 10018-3363.)

JOINT INDUSTRIAL COUNCIL (JIC)

JIC H-1 - Hydraulic Standards for Industrial Equipment and General Purpose Machine Tools.

(Application for copies should be addressed to the Joint Industrial Council, c/o The National Machine Tool Builders' Association, 7901 Westpark Drive, McLean, VA 22101-4269.)

NATIONAL ELECTRICAL MANUFACTURERS' ASSOCIATION (NEMA)

NEMA MG 1 - Motors and Generators. (DOD adopted)

(Application for copies should be addressed to the National Electrical Manufacturers' Association, 2101 L Street, N.W., Washington, DC 20037-1580.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), one complete machine shall be subjected to first article inspection (see 6.4) in accordance with 4.4.

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3.2 Design. The machine shall be new (not a prototype) and one of the manufacturer's current production models capable of operations in accordance with the requirements herein. The machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be readily accessible for adjustment, replacement, or repair.

3.2.1 Measuring and indicating device calibrations. Measuring and indicating devices shall be graduated in inch-pound (see 6.5.1), metric (see 6.5.2), or dual (inch-pound and metric) units as specified by the procuring activity (see 6.2). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

3.2.1.1 Dials. All dials used to indicate machine axes position or tool movement shall be graduated in increments not larger than .015 inch. Dial and handwheel circumferences shall be permanently and legibly engraved or etched on a nonglare background with graduations that can be read from the operator's normal position. All feed dials shall have independent zero adjustments and shall be calibrated in such a manner that the last dial graduation progresses into and is continuous with the first dial graduation as the dial is rotated through the zero position.

3.2.2 Reclaimed materials. The machine may contain reclaimed materials to the maximum extent possible provided such materials will not jeopardize the intended use, performance, or design life of the machine. Reclaimed materials shall have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used, rebuilt, or remanufactured end products are allowed under this specification.

3.2.3 Energy efficiency. The drilling machine and its components that directly consume energy in normal operation shall be designed and constructed for energy efficiency in accordance with the developments available within the industry.

3.2.4 Controls. All mechanical, hydraulic, and pneumatic operating controls shall be located convenient to the operator's work station(s).

3.2.5 Safety and health requirements. All parts, components, mechanisms, and assemblies furnished on the drilling machine whether or not specifically required herein, shall conform to all requirements of OSHA 29 CFR 1910 and ANSI B11.8. If a conflict arises between 29 CFR 1910 and ANSI B11.8, 29 CFR 1910 shall apply. Additional safety and health requirements shall be as specified (see 6.2). Covers, guards, or other safety devices normally furnished as standard on the manufacturer's commercial drilling machine supplied to the commercial market shall be provided for the point of operation and all other parts of the machine that present safety hazards. If point-of-operation guarding is influenced by workpiece size and configuration, the procuring activity

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shall provide part design data to the supplier of the machine (see 6.2). The supplier of the machine shall provide the special guarding which shall meet OSHA 29 CFR 1910.

3.2.6 Mercury restriction. The drilling machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the machine.

3.2.8 Environmental protection. During the manufacture, operation, service, transportation, or storage of the machine and its components, materials hazardous to the ecological system as prohibited by applicable Federal, state, or local statutes in effect on the date of the contract shall not be used or emitted.

3.2.9 Lubrication. Means shall be provided to ensure sufficient lubrication for all moving parts. Recirculating lubrication systems shall include a cleanable or replaceable filter. Means for determining oil level in each reservoir shall be by an oil level sight gauge, a low oil level warning light, or similar device and shall be in clear view of the operator's normal work station. An automatic lubrication system shall be provided for all ways and axis screws. All oil-holes, grease fittings, and filler caps shall be readily accessible.

3.2.10 Interchangeability. All parts of the end item shall be manufactured to definite standards and tolerances that will provide for the interchangeability of respective parts between end items of the same model. All replacement parts shall be interchangeable with their respective original parts, and fit/function with mating components.

3.3 Construction. The drilling machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall withstand all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion or failure.

3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale, and mismatching. No processes such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Fastening devices. All fasteners shall be installed to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be permanently installed.

3.3.3 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

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3.3.4 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. Any material used for such operations shall be thoroughly removed from the part(s) upon completion of the operations.

3.3.5 Painting. Unless otherwise specified (see 6.2), the paint color and application shall be in accordance with the manufacturer's commercial practice.

3.3.6 Threads. All threaded parts used on the drilling machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Drive train gears. All gears of spindle and axis drive trains shall conform to or exceed all provisions of AGMA 390.03a, AGMA 218.01, and AGMA 2000-A88 for the inch-pound system of measurement or ANSI/ISO 54 for the metric system of measurement. The gears shall be manufactured of a suitable ferrous material. The gears shall be of proper width and size to transmit full-rated torque and horsepower throughout the speed ranges without failure for the expected service life of the machine. Gears in the drive train shall be hardened and ground steel. Working surface hardness shall be not less than Rockwell C-48.

3.3.8 Electromagnetic interference control. When specified (see 6.2), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystem(s) class and the emission and susceptibility requirements shall be as specified.

3.4 Components. The drilling machine shall consist of not less than the following components.

3.4.1 Base. The base shall be a one-piece casting. The top of the base shall be machined for mounting the column and to provide a work surface. The work surface shall have not less than three, 5/8 inch or larger T-slots. T-slots shall be in accordance with ANSI B5.1. A coolant fluid trough shall extend around the periphery of the work surface and be connected to a coolant reservoir which shall be integral with the base. A screen to prevent chips from entering the reservoir shall be furnished.

3.4.2 Plain type table. When specified (see 6.2), the machine shall have a plain type table configured and machined for attachment to the work surface of the base. The plain table shall be of box construction with finished top and side working surfaces. The top surface shall have three or more T-slots and the side surface two or more T-slots. T-slots shall be in accordance with ANSI B5.1.

3.4.2.1 Universal tilting type table. When specified (see 6.2), the machine shall have universal tilting type table configured and machined for attachment to the work surface of the base. The table shall consist of a base on which shall be mounted a tilting work table with two machine finished work surfaces at 90° to each other. The top surface of the table shall be not less than 20 inches by 20 inches. The side work surface of the table shall be not less than 10 inches by 20 inches. The table surfaces shall be machined and provided with T-slots, three or more on the

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top and two or more on the side. T-slots shall be in accordance with ANSI B5.1. The table shall be arranged so that either work surface can be set in a vertical position by means of a worm and segment gear and have means for securing the work surfaces in position. The segment shall be graduated in degrees to indicate the angle at which the table is set. The base of the table shall be finished and provided with slots or bolt holes.

3.4.3 Column. The column shall consist of two ferrous castings, tubular in construction, with an inner member or trunk and an outer member or sleeve. The trunk shall be reinforced by internal ribs and shall have a flange for mounting on the base. The outer sleeve shall rotate 360° around the trunk on ball or roller bearings. The column shall be arranged with a clamping mechanism for securing the trunk to the sleeve binding the outer sleeve to the trunk with a contract area of 360°. The finish on the surface giving support to the arm shall not be rougher than 32 rms.

3.4.4 Arm. The arm shall be a reinforced ferrous casting, machined and fitted to the column. The length of the arm bearing on the column shall be not less than twice the diameter of the column. Means shall be provided for lubricating the column sleeve when the arm is raised and lowered. Wipers shall be provided for cleaning the column sleeve. Ways shall be provided for movement of the head. The ways which provide the main support for the head shall have a surface hardness of not less than Rockwell C-50. The ways may be of the replaceable strip type. The arm shall have a rack gear for traversing the head along the arm. The arm shall rotate through 360° by means of the column sleeve rotation. Vertical arm movement shall be accomplished by an elevating screw and nut assembly driven either separately or by the main drive motor. The nut assembly shall be mounted on ball or roller bearings and shall be equipped with a safety device that supports the arm in case of thread stripping. Automatic limit stops shall be provided to arrest the travel of the arm at both ends of the vertical range. The elevating screw shall be supported at both ends with support brackets and bearings and affixed to the column at both top and bottom. A clamping device shall be furnished for securing the arm to the column. When specified (see 6.2), machines with column diameters larger than 13 inches shall have either electric, hydraulic, or electrohydraulic clamping devices for securing the arm to the column and the column trunk to the column sleeve.

3.4.5 Head. The head shall be fully enclosed, machined and fitted to the arm. A clamp shall be provided for locking the head to the arm. The head shall be traversed along the arm by means of a power or manually operated rack and pinion gear mechanism. When specified (see 6.2), the heads of machines with column diameters larger than 17 inches shall be traversed by means of a power operated mechanism. When power traverse is provided, the head shall incorporate and interlock or slip clutch to prevent damage from engagement of the rapid traverse while the head is in a locked position. The spindle and all rotating member sliding surface sleeves, bearings and component shall be lubricated by a constant supply of oil during use.

3.4.6 Spindle. The spindle shall be of heat-treated alloy steel with a surface hardness of not less than Rockwell C-40. The spindle shall be mounted in a sleeve which shall support or drive the spindle. The spindle

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shall be supported by ball or roller bearings. The spindle counterbalance mechanism shall be fully enclosed in the head. The spindle taper shall be of the size specified in table I, and shall comply with the requirements of ANSI B5.10.

3.4.7 Speeds and feeds. The spindle speed and feed change controls shall be provided on the head. Speed changes shall be controlled by not more than three conveniently located levers and accomplished through sliding gears and positive tooth clutches. All speed and feed gears shall be of heat-treated alloy steel. Sliding gears shall slide on multiple hardened and ground spined shafts having not less than three splines with integral keys. A depth gage shall be provided which shall automatically disengage the power feed at any predetermined depth indicated on the graduated dial. A device shall be included to automatically disengage the feed when the limit of spindle travel is reached. The feed engagement shall be operated by a positive tool or friction type clutch with means for adjustment.

3.4.8 Coolant system. The coolant system shall include a sump or reservoir, a power driven pump, and all necessary piping. The sump or reservoir shall have sufficient capacity to permit full flow of coolant. The system shall have means for draining and cleaning and shall include a baffle or a strainer which is easily removable for cleaning. Means shall be provided to permit the operator to direct and control the amount of coolant over the entire work area.

3.4.9 Hydraulic system. The hydraulic system shall conform to or exceed the requirements of JIC H-1. The hydraulic system shall be complete, including all pumps, valves, piping, cylinders, and pressure controls. Overpressure protection shall be provided in the high pressure line to prevent damage to components. A filter system shall be provided to ensure delivery of clean fluid. The hydraulic reservoir shall have means for determining fluid level, provisions for draining, and a cleanout access plate if the reservoir is not removable.

3.4.10 Electrical requirements. The machine shall be wired to conform to NFPA 79. Unless otherwise specified (see 6.2), each machine shall be designed to operate on 230/460-volts, 3-phase, 60 Hertz (Hz). Unless otherwise specified (see 6.2), the machine shall be initially wired for operation on 460 volts.

3.4.10.1 Motors. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed and permanently lubricated type. Unless otherwise specified (see 6.2), alternating-current (AC) motors shall be designed to operate on 60-Hz. The motor enclosure shall meet the requirements for a (dripproof, totally enclosed fan-cooled) enclosure. All motors shall conform to the requirements of NEMA MG-1.

3.4.10.2 Worklight. When specified (see 6.2), a worklight shall be provided. The light shall have a protective shield, an on-off switch, and shall be adjustable to reflect maximum light on the work area. If a built-in lighting system is provided, the adjustable light described above shall not be required.

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3.4.10.3 Hour meter. When specified (see 6.2), the drilling machine shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time of the main drive motor. The meter shall be of the nonresetting type and shall have a range of 0 to 99,999 hours in increments of not greater than 1 hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the machine.

3.5 Size and capacity. Unless otherwise specified (see 6.2), the size and capacity of the drilling machine shall meet the requirements of table I for the machine specified. Machine capacity shall be not less than the stated requirements of the size ordered. When a range is given, the required performance shall be from the stated minimum or less to the stated maximum or greater.

3.5.1 Physical size limitations. When specified (see 6.2), the physical size of the machine shall not exceed the height, width, and length restrictions specified by the procuring activity to ensure the machine will fit in its future operating location.

3.6 Alignment tolerances. Each drilling machine shall meet the alignment tolerances specified in table II.

3.7 Performance. The drilling machine shall be capable of performing drilling, tapping, boring, and reaming operations for accomplishing all functional characteristics for which it is designed along with those required by table I. The machine shall also be capable of performing full horsepower drilling without any evidence of chatter, stalling, or permanent distortion.

3.8 Standard equipment. All standard equipment normally provided with the manufacturer's commercial machine shall be furnished.

3.9 Optional equipment. Optional equipment specified in the following subparagraph shall be furnished with each machine, as specified (see 6.2). A tapping attachment, unless a tapping feature is built into the machine.

3.10 Additional requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified (see 6.2).

3.11 Repair parts. Required repair parts shall be furnished as specified (see 6.2).

3.12 Marking on instruments, control panels, charts, and plates. All words on instruments, control panels, charts, and plates shall be in the English language. Characters shall be permanently marked in boldface on a contrasting background. All plates shall be corrosion resistant.

3.12.1 Lubrication plate. When specified (see 6.2), a lubrication plate shall be attached to each machine. Unless otherwise specified (see 6.2), the information provided on the plate shall be as listed below.

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Points of lubricant application
Servicing interval
Type of lubricant
Federal or military specification for each lubricant (if aplicable)

3.12.2 Nameplate. A nameplate shall be securely attached to each machine. Unless otherwise specified (see 6.2), the nameplate shall contain the information listed below.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Amp rating of largest motor
Short circuit/over-current protection rating
Contract number or order number
National Stock Number or Plant Equipment Code
Date of manufacture

3.12.3 Hydraulic fluid requirements plate. When specified (see 6.2), a hydraulic fluid requirements plate shall be attached to the machine at the hydraulic reservoir fill point. Unless otherwise specified (see 6.2), the plate shall contain the following information.

Type of fluid (SAE number or military or Federal specification number)
Type of filter

3.13 Workmanship. Workmanship of the drilling machine and its accessories shall meet all requirements specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize

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submissions of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the test conditions specified in 4.7.1.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.6 and all tests in 4.7. The machine shall pass the examination and all tests to be accepted.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each drilling machine prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of the examination in 4.6, the tests in 4.7, and the inspection in 4.8. The machine shall pass the examination, all tests, and the inspection to be accepted.

4.6 Examination. Each drilling machine shall be examined to determine compliance with all requirements of this specification.

4.7 Tests.

4.7.1 Test conditions. All tests shall be performed in an indoor facility with ambient conditions of 41^{0} to 104^{0} Fahrenheit and 20 to 95 percent relative humidity.

4.7.2 Operational test. Each drilling machine shall be operated at no load for not less than 30 minutes. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

4.7.3 Alignment tolerance test. Each machine shall be tested to determine compliance with the alignment tolerance requirements of 3.6. The alignment tolerance test shall be performed prior to and subsequent to the performance test(s) given in 4.7.4. Any machine adjustments required to comply with 3.6 during or after the performance test shall be cause for rejection.

4.7.4 Performance tests. Each machine shall drill 15 holes, of maximum rated size for the machine in 1-inch thick soft cast iron. For this test drill size shall be 2 inches for 9-inch column machines, 2-1/4 inches for 11-inch column machines and 4 inches for all other column diameter machines covered by this specification. Drilling time shall be not more

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than 20 minutes, total elapsed time. There shall be no evidence of overloading, overheating, excessive vibration, or deformation of any part of the machine during or after drilling operation. All performance tests shall be completed without making any adjustments or performing any maintenance on the machine.

4.7.5 Electromagnetic interference control test. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.8 using the procedures given in MIL-STD-462.

4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-18058.

6. NOTES

6.1 Intended use. The radial drilling machines covered by this specification are intended for drilling, reaming, spot facing, tapping, boring and counterboring operations on work too large or too heavy to be placed on an upright drilling machine.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type, class, and size required (see 1.2).

c. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

d. First article, if required (see 3.1).

e. System of units for measuring and indicating devices (see 3.2.1).

f. Specify configuration of additional point-of-operation guarding, if required (see 3.2.5).

g. Painting, if different (see 3.3.5).

h. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.8).

i. Plain table, if required (see 3.4.2).

j. Universal tilting table, if required (see 3.4.2.1).

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k. Power clamping device, if required (see 3.4.4).

l. Type of traversing mechanism, if different (see 3.4.5).

m. Electrical requirements, if different (see 3.4.10).

n. Initial voltage, if different (see 3.4.10).

o. Motors, if different (see 3.4.10.1).

p. Worklight, if required (see 3.4.10.2).

q. Hour meter, if required (see 3.4.10.3).

r. Size and capacity, if different (see 3.5).

s. Specify physical size limitation, if required (see 3.5.1).

t. Specify and fully describe optional equipment, if required (see 3.9).

u. Additional requirements, if required (see 3.10).

v. Specify applicable repair parts, if required (see 3.11).

w. Lubrication plate, if required (see 3.12.1).

x. Information to be provided on lubrication plate, if different (see 3.12.1).

y. Nameplate, if different (see 3.12.2).

z. Hydraulic plate, if required (see 3.12.3).

aa. Inspection conditions, if different (see 4.3).

bb. First article inspection, if different (see 4.4).

cc. Quality conformance inspection, if different (see 4.5).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

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Reference Paragraph	DID Number	DID Title	Suggested Tailoring
3.13	DI-TMSS-80527	Commercial Off-The-Shelf (COTS) Manuals.	Require all data be in the English language.
3.13	DI-TMSS-80528	Supplemental Data for Commercial Off-The-Shelf (COTS) Manuals.	Require all data be in the English language.

The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.3.1 Technical data. It should be specified in the "Remarks" section of the DD Form 1423 that all technical data ordered shall be written in the English language.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractor's current inventory and the number of items to be tested as specified in 4.4. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Definitions.

6.5.1 Inch-pound units. Inch-pound units are a system of measures based on the yard and pound commonly used in the United States of America and defined by the National Institute of Standards and Technology. Inch-pound units having the same names in other countries may differ in magnitude.

6.5.2 Metric units. Metric units are a system of basic measures defined by the International System of Units based on "Le Systeme International D'Unites (SI)," of the International Bureau of Weights and Measures. These units are described in ASTM E 380 and IEEE 268 (MIL-STD-961).

6.6 Measurement system. In this specification, all measurements, dimensions, sizes, and capacities are given in inch-pound units. These measurements may be converted to metric units through the use of the conversion factors and methods specified in FED-STD-376.

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6.7 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

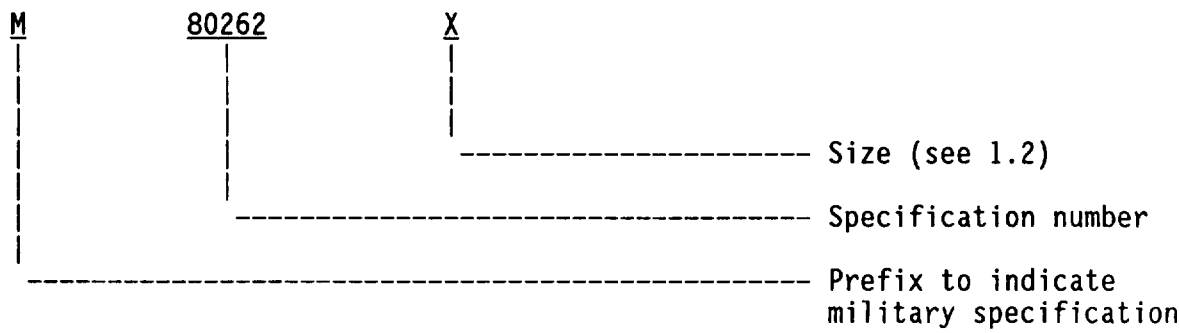
6.9 Safety and health requirements. OSHA 29 CFR 1910 limits only the total hazard level (noise, radiation, electromagnetic emissions, noxious vapors, air contaminants, and heat) of the environment in which a machine will operate. It does not limit the hazard level of individual machines in an operating environment. The procuring activity should analyze the existing hazard level in the proposed operating environment and specify additional requirements necessary to integrate this new machine into its future environment.

6.9 Training. Training required by the procuring activity should be provided as specified in the contract.

6.10 Warranty. Warranty requirements should be as specified by the procuring activity in the contract.

6.11 Inspection location. The contractor should identify in his response to the solicitation the location where inspection and tests are to be performed.

6.12 Part or Identifying Number (PIN). The PIN to be used for machines acquired to this specification are created as follows:



6.14 Subject term (keyword) listing.

Boring
Drilling
Facing
Heavy duty
Reaming
Tapping

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TABLE I. Sizes and capacities

Characteristics	Size A4	Size A6	Size B1	Size B2	Size B3
Column diameter, inches	9	9	11	11	11
Arm length, feet	3	4	3	4	5
Spindle to base, maximum, inches	53	53	55	55	55
Spindle to base, minimum, not more than, inches	14	14	14	14	14
Arm bearing length on column, inches	18	18	22	22	22
Arm traverse on column, inches	26	26	25	25	25
Spindle traverse in head, inches	10	10	12	12	12
Head traverse on arm, inches	25	37	25	37	49
Spindle center to column, minimum, not more than, inches	13	13	11	11	11
Base height, inches	6	6	8	8	8
Base working length, inches	41	53	41	53	65
Base working width, inches	30	30	36	36	36
Spindle speeds, number of	9	9	12	12	12
Spindle speed range, RPM	100 to 1500	100 to 1500	80 to 1500	80 to 1500	80 to 1500
Spindle feeds, number of	4	9	6	6	6
Spindle feed range, inch per revolution	.004 to .020	.004 to .020	.003 to .020	.003 to .020	.003 to .020
Spindle diameter at point of drive, inches minimum	1-3/8	1-3/8	1-1/2	1-1/2	1-1/2
Motor horsepower	5	5	5	5	5
Taper in spindle, morse No:	4	4	4	4	4

NOTE: Machine sizes and capacities shall not be less than the stated requirements of the size ordered, nor greater than the next larger size shown in table I. When a range is shown, the required performance is from the stated minimum or less to the stated maximum or greater.

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TABLE I. cont. Sizes and capacities

Characteristics	Size C1	Size C2	Size D1	Size D2	Size D3
Column diameter, inches	13	13	15	15	15
Arm length, feet	4	5	4	5	6
Spindle to base, maximum, inches	60	60	66	66	66
Spindle to base, minimum, not more than, inches	11	11	11	11	11
Arm bearing length on column, inches	26	26	30	30	30
Arm traverse on column, inches	28	29	35	35	35
Spindle traverse in head, inches	18	18	18	18	18
Head traverse on arm, inches	32	44	32	44	56
Spindle center to column minimum, not more than, inches	16	16	16	16	16
Base height, inches	8	8	8	8	8
Base working length, inches	53	65	53	65	77
Base working width, inches	36	36	40	40	40
Spindle speeds, number of	32	32	32	32	32
Spindle speed range, RPM	15 to 1200	15 to 1200	15 to 120	15 to 1200	15 to 1200
Spindle feeds, number of	16	16	16	16	16
Spindle feed range, inch per revolution	.004 to .125	.004 to .125	.004 to .125	.004 to .125	.004 to .125
Spindle diameter at point of drive, inches minimum	2-1/2	2-1/2	2-3/16	2-3/16	2-3/16
Motor horsepower	10	10	15	15	15
Taper in spindle, morse No:	5	5	5	5	5

NOTE: Machine sizes and capacities shall not be less than the stated requirements of the size ordered, nor greater than the next larger size shown in table I. When a range is shown, the required performance is from the stated minimum or less to the stated maximum or greater.

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TABLE I. cont. Sizes and capacities

Characteristics	Size E1	Size E2	Size E3	Size F1	Size F2
Column diameter, inches	17	17	17	19	19
Arm length, feet	5	6	7	6	7
Spindle to base, maximum, inches	72	72	72	78	78
Spindle to base, minimum, not more than, inches	15	15	15	15	15
Arm bearing length on column, inches	34	34	34	38	38
Arm traverse on column, inches	37	37	37	43	43
Spindle traverse in head, inches	18	18	18	18	18
Head traverse on arm, inches	47	58	70	57	79
Spindle center to column, minimum, not more than, inches	15	15	15	15	15
Base height, inches	10	10	10	10	10
Base working length, inches	65	77	89	77	89
Base working width, inches	50	50	50	50	50
Spindle speeds, number of	32	32	32	32	32
Spindle speed range, RPM	15 to 1200	15 to 1200	15 to 1200	15 to 1200	15 to 1200
Spindle feeds, number of	16	16	16	16	16
Spindle feed range, inch per revolution	.004 to .125	.004 to .125	.004 to .125	.004 to .125	.004 to .125
Spindle diameter at point of drive, inches minimum	2-1/2	2-1/2	2-1/2	2-1/2	2-1/2
Motor horsepower	15	15	15	20	20
Taper in spindle, morse No:	5	5	5	6	6
Tap leads, threads per inch: 1	8	8	8	8	8
2	11-1/2	11-1/2	11-1/2	11-1/2	11-1/2
3	14	14	14	14	14

NOTE: Machine sizes and capacities shall not be less than the stated requirements of the size ordered, nor greater than the next larger size shown in table I. When a range is shown, the required performance is from the stated minimum or less to the stated maximum or greater.

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TABLE I. cont. Sizes and capacities

Characteristics	Size F3	Size G1	Size G2	Size G3	Size G4
Column diameter, inches	19	22	22	22	22
Arm length, feet	8	7	8	9	10
Spindle to base, maximum, inches	78	96	96	96	96
Spindle to base, minimum, not more than, inches	16	22	22	22	22
Arm bearing length on column, inches	38	44	44	44	44
Arm traverse on column, inches	43	50	50	50	50
Spindle traverse in head, inches	18	24	24	24	24
Head traverse on arm, inches	79	67	79	91	103
Spindle center to column, minimum, not more than, inches	15 - 17	17	17	17	17
Base height, inches	10	12	12	12	12
Base working length, inches	101	90	102	114	126
Base working width, inches	50	57	57	57	57
Spindle speeds, number of	32	36	36	36	36
Spindle speed range, RPM	15 to 1200	8 to 800	8 to 800	8 to 800	8 to 800
Spindle feeds, number of	16	18	18	18	18
Spindle feed range, inch per revolution	.0025 to .125	.004 to .125	.004 to .125	.004 to .125	.004 to .125
Spindle diameter at point of drive, inches minimum	2-1/2	3	3	3	3
Motor horsepower	20	20	20	20	20
Taper in spindle, morse No:	6	6	6	6	6
Tap leads, threads per inch: 1	8	8	8	8	8
2	11-1/2	11-1/2	11-1/2	11-1/2	11-1/2
3	14	14	14	14	14

NOTE: Machine sizes and capacities shall not be less than the stated requirements of the size ordered, nor greater than the next larger size shown in table I. When a range is shown, the required performance is from the stated minimum or less to the stated maximum or greater.

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TABLE I. cont. Sizes and capacities

Characteristics	Size H1	Size H2	Size H3	Size H4	Size H5
Column diameter, inches	26	26	26	26	26
Arm length, feet	8	9	10	11	12
Spindle to base, maximum, inches	100	100	100	100	100
Spindle to base, minimum, not more than, inches	22	22	22	22	22
Arm bearing length on column, inches	52	52	52	52	52
Arm traverse on column, inches	50	50	50	50	50
Spindle traverse in head, inches	24	24	24	24	24
Head traverse on arm, inches	80	92	104	116	128
Spindle center to column, minimum, not more than, inches	22	22	22	22	22
Base height, inches	12	12	12	12	12
Base working length, inches	102	114	126	138	150
Base working width, inches	57	57	57	57	57
Spindle speeds, number of	36	36	36	36	36
Spindle speed range, RPM	8-800	8-800	8-800	8-800	8-800
Spindle feeds, number of	18	18	18	18	18
Spindle feed range, inch per revolution	.004 to .125	.004 to .125	.004 to .125	.004 to .125	.004 to .125
Spindle diameter at point of drive, inches minimum	3	3	3	3	3
Motor horsepower	20	20	20	20	20
Taper in spindle, morse No:	6	6	6	6	6
Tap leads, threads per inch: 1	8	8	8	8	8
2	11-1/2	11-1/2	11-1/2	11-1/2	11-1/2
3	14	14	14	14	14

NOTE: Machine sizes and capacities shall not be less than the stated requirements of the size ordered, nor greater than the next larger size shown in table I. When a range is shown, the required performance is from the stated minimum or less to the stated maximum or greater.

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TABLE II. Alignment and accuracy tolerances.

Components	Maximum Allowance Deviation
Work surface, flat in longitudinal direction	0.0012 inch per foot
Work surface, flat in traverse direction traverse column 1	0.0012 inch per foot
Column square with base in plane through center line of base	
For machines up to 4 foot 7 inch length radial arm	0 to 0.0024 inch per foot
For machines over 4 foot 7 inch length of arm	0 to 0.0036 inch per foot
Column square with base in plane perpendicular to plane through center line of baseplate	0.0012 inch per foot
Arm parallel with base (low at outer end)	0 to 0.0024 inch per foot
For machines up to 4 foot 7 inch length of radial arm	0 to 0.0036 inch per foot
For machines over 4 foot 7 inch length of radial arm	0 to 0.0024 inch per foot
Spindle runout (using 12 inch long mandrel 12" from spindle nose	0.002 inch
Spindle square with base in plane through center line of base, (lower end inclined toward column)	0 to 0.0024 inch per foot
Spindle square with work surface of base in plane 90 degrees to above plane	0.0012 inch per foot
Travel of spindle sleeve square with work surface of base in plane through center line of base (inclined at lower end toward column)	0 to 0.004 inch per foot
Travel of spindle square with base in plane 90 degrees to above plane	0.002 inch per foot
* Drill spindle square with plain type table mounted on base in plane through center line of base (spindle inclined at lower end toward column)	0 to 0.0024 inch per foot
* Drill spinDle square with plain type table mounted on base in plane perpendicular to above plane	0.0012 inch per foot

When testing, the arm shall be positioned half-way up the column and the head half-way along the arm.
* Indicates when a plain type table is furnished.

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Custodians:
Army - AL
Navy - SH
Air Force - 99

Review activity:
Air Force - 84

User activities:
Army - SM
Navy - YD, MC

Preparing activity:
DLA - IP

Project (3413-0146)

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-D-80262B	2. DOCUMENT DATE (YYMMDD) 90-03-26

3. DOCUMENT TITLE
Drilling Machine, Radial, Floor Mounted

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Defense Industrial Plant Equipment Center	b. TELEPHONE (Include Area Code) (1) Commercial 901-775-6257 (2) AUTOVON 683-6257
c. ADDRESS (Include Zip Code) 2163 Airways Boulevard Memphis, Tennessee 38114-5051	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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